20 Years Evolution of Electrothermal MEMS Mirror for EndoOCT

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1 Objective of this document

If someone googles about endoscopic OCT based on electrothermal actuation, chances are, you will notice the name Huikai Xie. Well, it's not by chance. Prof Xie HuiKai & his group has been working on electrothermal endoOCT for nearly 20 years, starting from 2001.

Over the past 20 years, they improvised their design a lot, to the point where there are relatively very few papers from other groups. Thus, if one wants to push the boundaries further. it's imperative to understand their design evolution.

As mentioned in the PhD thesis [19] of one of his student who graduated in 2018. Their work can be classified as

Generation	Type and color coding
First generations	A1 A2 A3 A4 A5 Curved
Second generations	simple ISC double ISC triple ISC
Third generations	LSF
Fourth generations	torsion bar

Figure 1: Classification From: [19]

2 Thermal Actuation Basics

A thermal bimorph beam is composed of two material films, with different coefficients of thermal expansion (CTE), bonded at an interface. Typically thermal bimorphs are made of one material with a **low CTE, such as a dielectric like SiO2**, and another material with a **high CTE, such as metal like Al**.

When the temperature of the bimorph is raised, the high CTE material will want to expand more than the low CTE material. Since both materials are bonded together, stress develops in both material layers due to the bonded interface constraint. The high CTE material exhibits a compressive stress because it is stretched below its equilibrium length, and the low CTE material exhibits a tensile stress because it is stretched past its equilibrium length. The stresses that develop upon an increase in temperature will cause the bimorph to curl towards the low CTE material to minimize the internal energy stored by the stress.

At the temperature at which the bimorph materials are deposited, assuming no residual stress is present from processing conditions, both materials will be at their equilibrium lengths and the bimorph will be flat. The opposite effect happens when the temperature of a bimorph is lowered.

The amount of curl of a bimorph is measured by its radius of curvature, ρ. The radius of curvature of a bimorph depends both on stresses that evolve from a temperature change and residual stresses that may be present in the material layers. Residual stresses affect the bimorph curl in exactly the same way as stresses that develop from temperature [22].

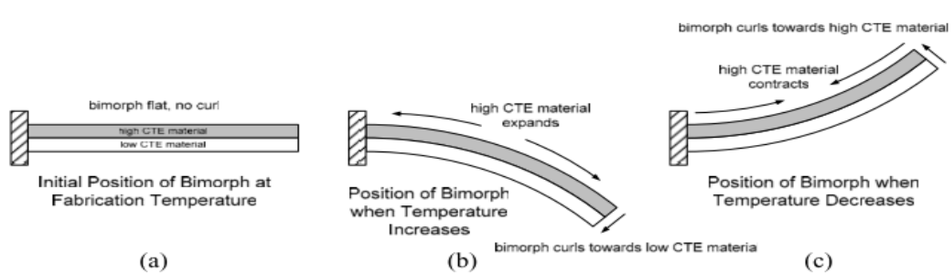

Figure 2: From: [22]

3 First Generation

Note: The first electorthermally actuated scanner was repoted in 1992 by Buser et al. [1]. But, we are ignoring it and other earlier works Bühler et al. [2], Jiang et al. [10], Sinclair [18].

3.1 A1

3.1.1 A1 (a)

Design: Pan et al. [15] work in 2001 is the ancestor of the modern electrothermally actuated micro-scanner. The work is described in detail in [22]. The micro-mirror has reflective top layer(Al), a single crystal silicon (SCS) membrane bottom layer which adds stiffness to the mirror to ensure reasonable surface flatness, and an SiO_2 layer in between the Al and Si layers. The bimorph actuator is composed of an Al top layer, an SiO_2 bottom layer, and a polysilicon resistive heater layer that is embedded in the SiO_2 layer. As shown below.

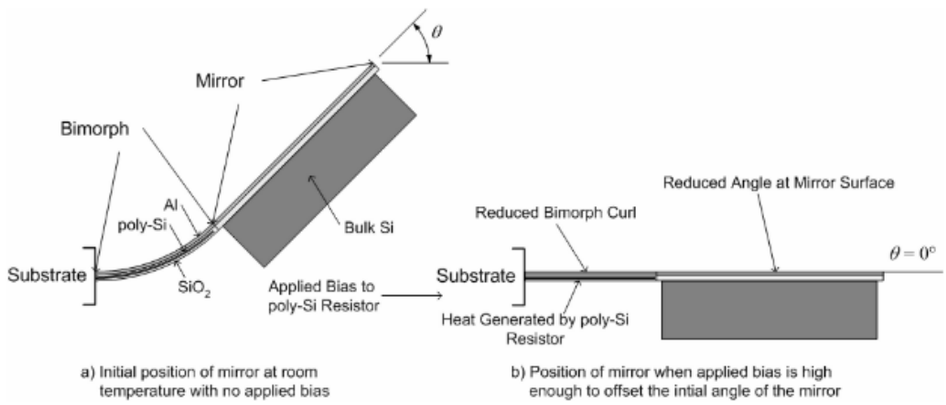

Figure 3: From: [22]

Working: When a voltage is applied to the polysilicon resistor, the resistor dissipates power and causes the bimorph temperature to increase. Notice that the bimorph temperature can only be raised and not lowered by the polysilicon resistor, hence the mirror can rotate only in one direction with respect to its initial angular position. If one wishes to operate the mirror in two rotation directions, then an initial bias point is needed to lower the angle of the mirror to a new initial point. After that the mirror can be rotated in both directions by either increasing or decreasing the voltage to the polysilicon resistor.

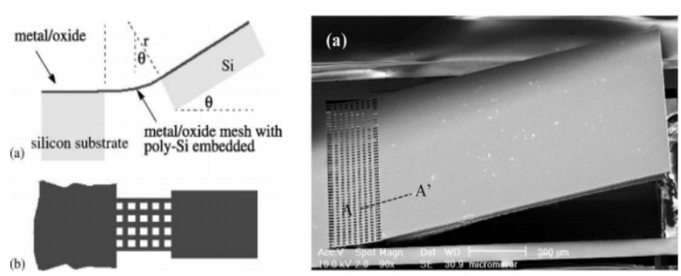

Figure 4: A1 (a) From: [25]

Limitations: However, the buckling of the employed bimorph mesh structure causes a discontinuity in the angle versus current curve, which limits the usable angle range to about 5 °[4].

The discontinuity is believed to be caused by a temperature induced stress along the width of the bimorph. The induced stress evolves because the connections of the bimorph mesh to the substrate and mirror plate constrain the bimorph mesh along its width, limiting the ability of the bimorph to expand laterally at an elevated temperature. As the actuation voltage and bimorph temperature increase, the lateral stress increases and eventually reaches a critical point where the bimorph buckles. At this critical point, the rotation angle of the mirror increases suddenly, causing the discontinuity in the angle-voltage curve [22, 25].

3.1.2 A1 (b)

To eliminate the angle-voltage discontinuity observed in the first design, a new mesh design [7] was created with a polysilicon resistor transversely oriented with respect to the substrate attachment point [7]. As shown below.

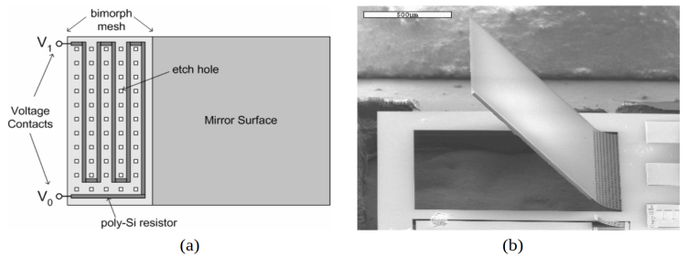

Figure 5: A1 (b) From: [22]

It was experimentally determined that the transverse arrangement of the polysilicon resistor provides a smooth actuation curve. The buckling phenomenon is not completely understood, so it is not exactly clear why the transverse arrangement of the polysilicon resistor eliminates the buckling. The transverse design also exhibited a greater initial angle and rotation angle versus voltage characteristic compared to the longitudinal design [22].

Unable download this paper: [7]

3.2 A2

Instead of using a continuous bimorph mesh that was used in the previous designs, the bimorph actuator was split into an **array of bimorph beams** [4], as shown below.

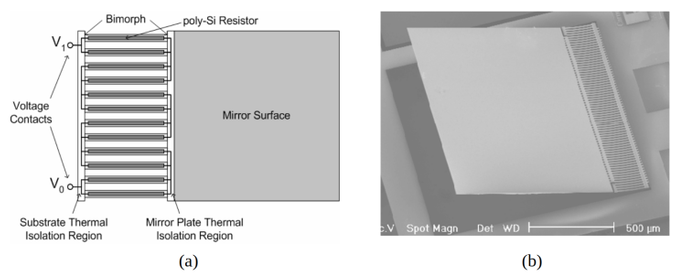

Figure 6: A1 (b) From: [22]

Since the width of a bimorph beam attachment point is much smaller than the width of the bimorph mesh attachment point, the bimorph beams are less constrained laterally by their fixed attachment points, thus a smaller lateral stress exists upon actuation.

Another difference in the new design is that **thermal isolation regions** were added to isolate the bimorph beam array from the substrate and mirror plate regions. The thermal isolation regions are useful for two things.

- First, the thermal isolation region between the bimorph array and the substrate increases the average temperature of the bimorph array for a given bias, yielding a greater angular response of the mirror.
- Second, the thermal isolation layer between the bimorph and the mirror plate lowers the heat flux between the two regions upon actuation, resulting in a faster thermal response time of the bimorph.

3.3 A3

A3 is extending A2 in two-axis by placing two array of bimorph beams perpendicular to each other.

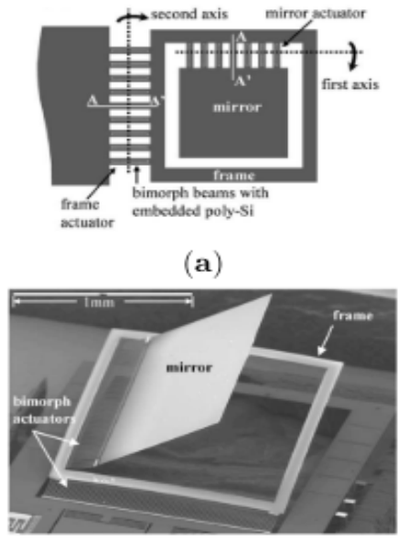

Figure 7: A3 From: [8]

Limitations: Unidirectional operation, non-stationary center of rotation, and large initial tilt angle of those micromirrors complicated the device packaging and optical design [8].

3.4 A4: Large Vertical Displacement (LVD)

In A4, the mirror surface is kept parallel to the substrate and also provides bi-directional scanning capability by using two complementarily-oriented electrothermal actuators. This is known as **large-vertical-displacement** (LVD) micromirror [8]. It can perform rotational scans, as well as generate large piston motion at low driving voltages. Out-of-plane displacement of the micromirror is provided by a pair of electrothermal actuators.

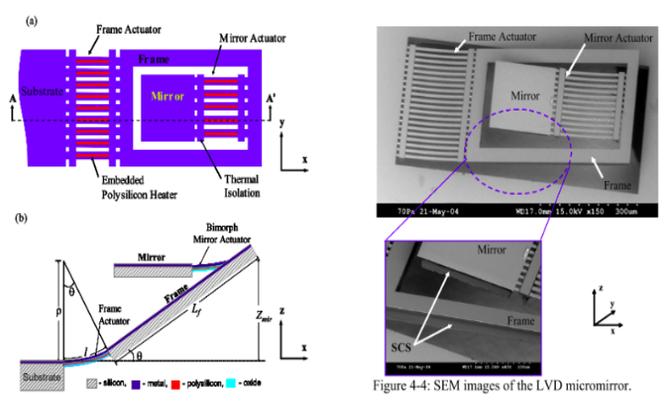

Figure 8: A4 From: [8]

3.5 A5

A5 is extending A4 in two-axis by placing two set of LVD beams perpendicular to each other

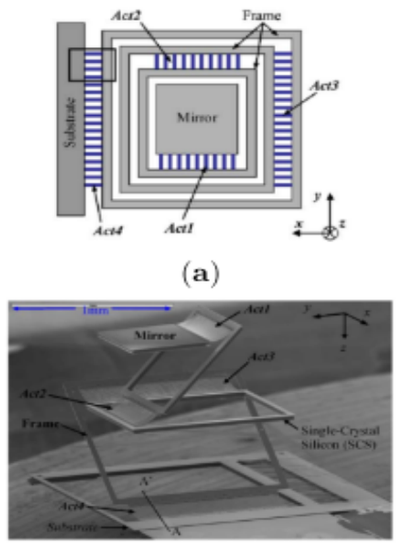

Figure 9: A5 From: [8]

Limitations: In this version, the leverages are even more pronounced than formerly so that this configuration cannot be utilized for a proper scanning for OCT as it also shows a strong thermal coupling between the frames. The consequent mass of the outermost frame limits the natural frequency of the overall device at 170 Hz [19].

3.6 Curved

Micromirror suffers a large center-shift during angular scanning as shown below. Such large center-shift hampers optical alignment.

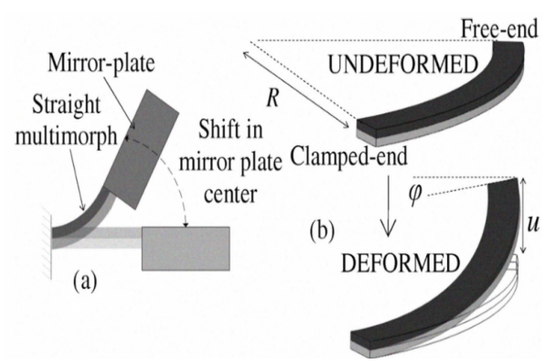

Figure 10: Curved From: [14]

A possible approach to compensate the large center-shift is to use actuators on opposite sides of the mirror plate [5], but the inclusion of additional actuators and the need for another set of actuators for 2D scanning further reduces the fill-factor and increases power consumption. Moreover, those micromirrors have large thermal time constants in the order of 10's milliseconds.

Thus, Pal and Xie [14] reported a **curved multimorphs** that undergo combined bending and twisting during deformation. The mirror-center shift due to bending and twisting partially compensate each other. This results in a low mirror center-shift.

Additionally, curved multimorphs achieve reasonable resonant frequency with a high fill factor whereas large number of straight multimorphs are required to achieve reasonably high resonant frequency resulting in a small fill-factor and high power consumption. Power consumption may be reduced by using thermal isolation at the ends of the actuator beams. However, the most commonly used thermal isolation material is SiO_2 which is brittle and increases susceptibility to impact failure [14].

Limitations: However, the center of the mirror is subject to strong fluctuations although Q-factors are not mentionned in these designs [19].

4 Second Generation

4.1 Issues in First Generation

Although the prior actuator designs solve the unidirectional scan and large initial tilt problems, they bring in a new lateral shift (L_s) during the piston motion of micromirror [23]. As shown below. This lateral shift distorts the microscopic imaging and reduces the effective optical aperture size [23].

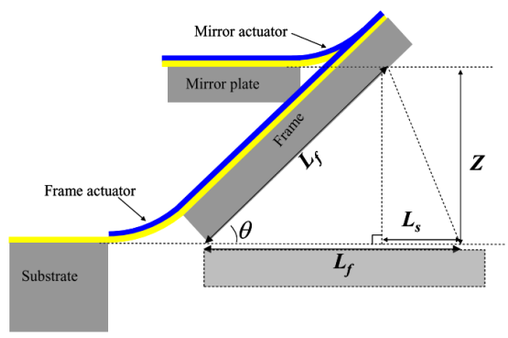

Figure 11: Lateral Shift (L_s) during Vertical Displacement(Z) From: [23]

Another disadvantage of the prior designs is that a certain ratio of the driving voltages for the two complementary actuators needs to be maintained in order to obtain an equal but opposite rotation angles of the two actuators thus a pure piston motion of the mirror plate without tilting, and this experimentally-determined ratio is not constant but increases with the applied voltages. This greatly complicates the driving of the device especially for the dynamic vertical scan.

Also, the fill-factor of the LVD micromirror is relatively low due to the large area taken by the frames and the bimorph beams.

Finally, the problem of non-stationary mirror center during actuation remains unsolved. As shown below.

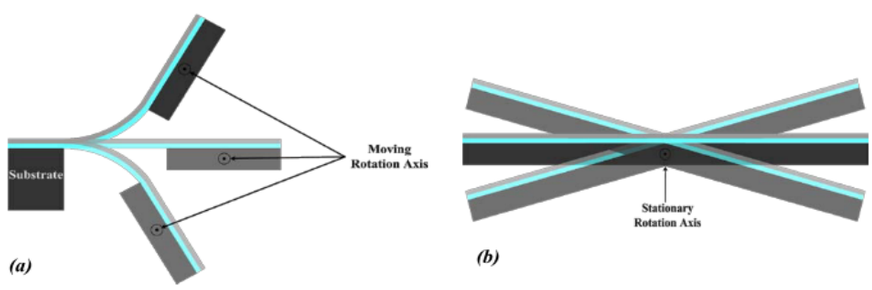

Figure 12: From: [9]

4.2 Inverted-Series-Connected (ISC)

4.2.1 Simple ISC (a)

If two bimorphs with inverted layer compositions are connected in series to form an S-shaped beam, as shown in Fig. 12b, the tilt angle at the tip of the second bimorph is compensated. However, the lateral shift still remains. If two S-shaped beams are connected in series with the second one folded, as in Fig. 12c, both the lateral shift and the tilt angle will be compensated.

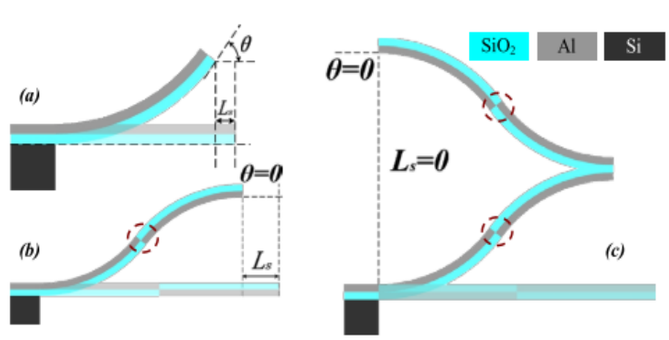

Figure 13: From: [9]

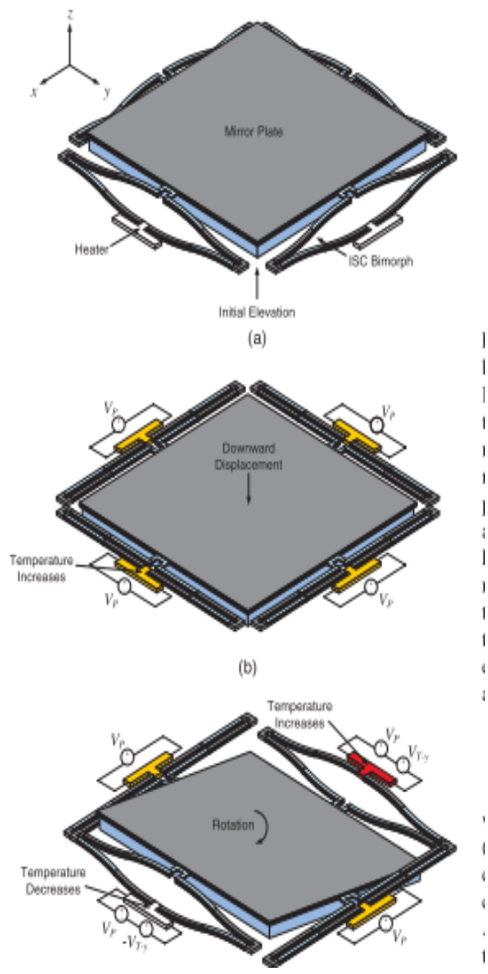

Figure 14: From: [21]

Limitations: (1) The mechanical robustness of the actuator is compromised by the weak connection between the bimorph segments, as indicated by the circled part in Fig. 12; (2) Instead of embedding the heater along the bimorph path, the device design has the heater placed at one end of the bimorph actuator, thus resulting in non-uniform temperature distribution and very slow thermal response time. (3) Pin holes in the SiO2 layer which resulted in unwanted etch to the poly-Si heater and caused open circuit [21].

4.2.2 Simple ISC (b)

An overlap portion is added at the middle of each S-shaped bimorph beam to increase the mechanical robustness. As shown in Fig. 14. Pt instead of Poly-Si is employed as the heater material and is embedded along the bimorph. Each pair of FDS actuators is connected to the mirror plate via a pure SiO2 joint which ensures good thermal isolation between actuators and mirror plate and keep the mirror plate from being heated up and deforms.

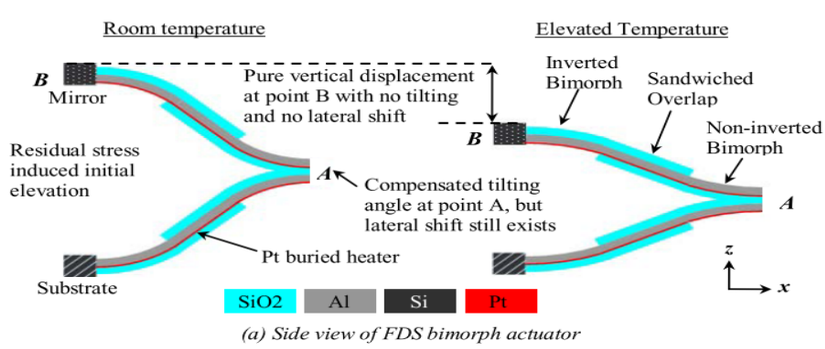

Figure 15: From: [9]

4.2.3 Hidden ISC

They improved the fill factor by hiding the actuator beneath the mirror as shown below in Fig.

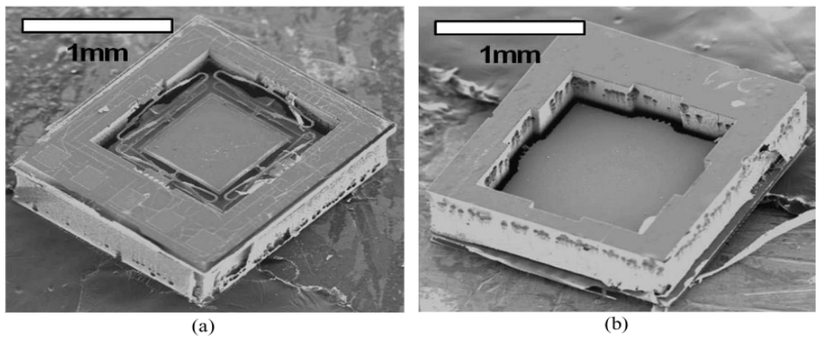

Figure 16: From: [9]

4.3 Double ISC

The vertical displacement was further improved by having double ISC as shown in Fig 16.

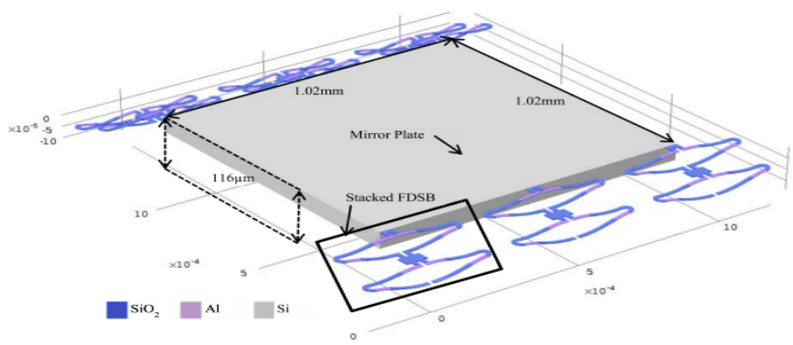

Figure 17: From: [17]

4.4 Triple ISC

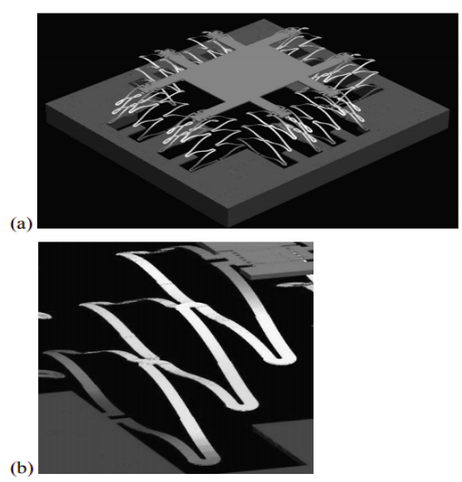

Figure 18: From: [24]

4.5 Gimbal-less Micromirror

It consists of a high reflective gold–chromium mirror plate, four flexural springs and four electrothermal bimorph actuator. As shown below:

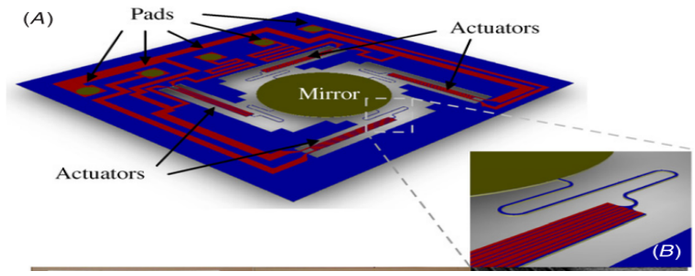

Figure 19: NUS; From: [26]

The work has inspired other similar works as described in [6, 11]

Limitations: The radius of curvature of the mirror surface is significantly low (about 5mm) due to the thin-film based mirror plate [9, 19] and may affect the depth of focus.

5 Third Generation

5.1 Lateral Shit Free (LSF)

The lateral shift in LVD is compensated here by having additional set of bimorph actuator. As shown below.

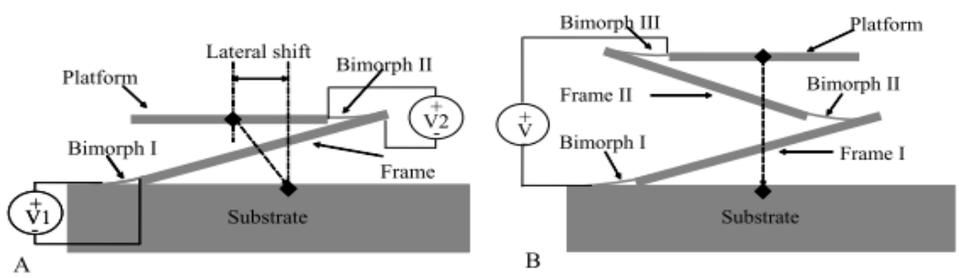

Figure 20: LVD vs LSF From: [23]

Also, It uses one uniform heater embedded along all actuators, so the driving is simplified and only one voltage is needed.

Limitations: It suffers from a slightly skewed scan pattern due to the structural asymmetry. The rotation-axis shifting, although small, still exists. Area efficiency of the LSF device is not high due to the large area taken by the bimorphs and frames. Lastly, the bulk silicon underneath the long frames also limits the fast response of the LSF device [9].

5.2 curved LSF

They improved area efficiency by using curved LSF actuators [12].

By using Curved LSF as shown in Fig 20.

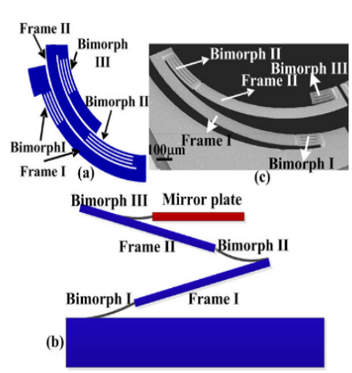

Fig. 2. (a) Top-view schematic of the CCBA. (b) Side-view schematic of the CCBA. (c) SEM of a fabricated CCBA.

Figure 21: From: [12]

5.3 Pretilted

By pre-tilting the mirror, they made the assembly process easier [3].

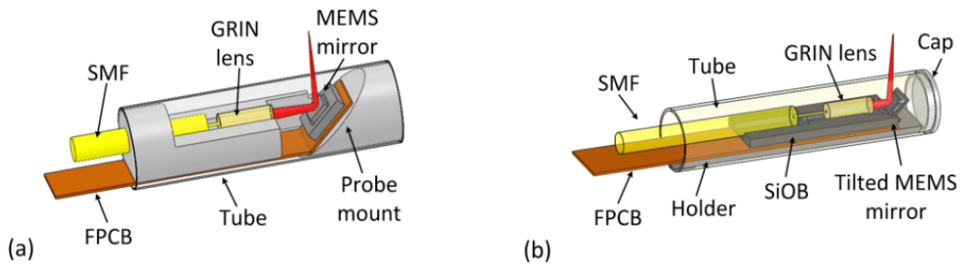

Figure 22: From: [3]

6 Fourth Generation

6.1 Torison Bar

Torsion bars are used to restrict the mirror plate motion to a pure rotation. ISC actuators are connected in series and parallel to optimizes the space around the mirror plate to increase the displacement and the force of the actuators without degrading the fill factor [20].

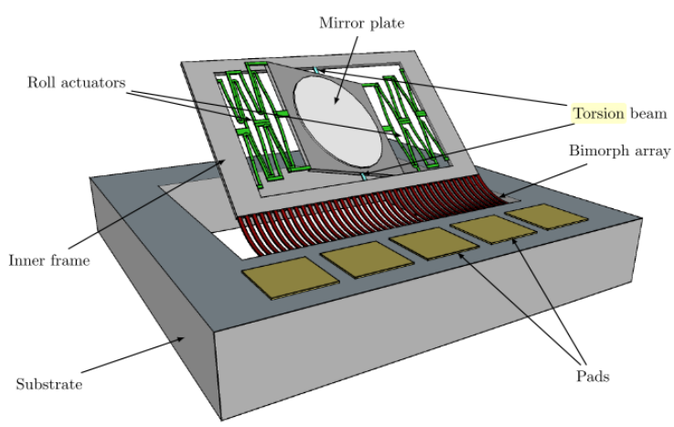

Figure 23: From: [20]

Limitations: The vibration of the torsion bar in other modes as shown in Fig 23 may affect the OCT performance.

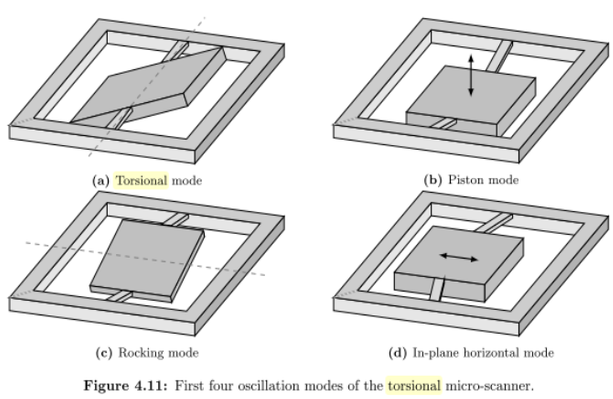

Figure 24: From: [19]

7 Why new is slow?

It is human nature to cautiously approach what is new, for it is foreign and untested. This cautious behavior translates to slow acceptance of new technologies and derivative products as they get introduced

into society. For example, demonstration of the first micromachined accelerometer took place in 1979 at Stanford University [16]. Yet it took nearly 15 years before it became accepted as a device of choice for automotive airbag safety systems. However, this psychological barrier appears to be declining [13].

8 Scope for Improvement

1. Sensing of Actuation w/o sensors (e.g: Back EMF)
2. Immersion in Liquid for increased FoV
3. Vaccum Packaging.
4. Integration of Laser Ablation + OCT
5. Hybrid (electrostatic + electrothermal)
6. Better Structural Design. (Four bar, Five bar)
7. Improvised Fabrication Method
8. Better Materials

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